Cpts 317. 10/19

Today: Context-free Grammar.

math def of c.f Gramar:

A c.f. grammar G is a tuple

$$\langle V, \Sigma, S, P \rangle$$

where: V: a fnite set of nonterminal symbols;

Σ: . . . . . - terminal symbols

$$(V \cap \Sigma = \emptyset)$$

$S \in V$ is the initial symbol;

P: is a fnite set grammar rules

(productions). Each rule is in the form:

$$A \longrightarrow \alpha$$

where $A \in V$, $\alpha \in (V \cup \Sigma)^*$.

Example. $S \longrightarrow 0S1 \mid \Lambda$

where: $V = \{S\}$,

$\Sigma = \{0, 1\}$

P has two grammar rules:

$$S \longrightarrow 0S1$$

$$S \longrightarrow \Lambda$$

Example.

$$\left.\begin{array}{l} S \longrightarrow AB \\ A \longrightarrow 0A1 \mid \Lambda \\ B \longrightarrow \Lambda \end{array}\right\} \text{4 rules}$$

$V = \{S, A, B\}$, $\Sigma = \{0, 1\}$.

We write $\alpha \Rightarrow_G^* \beta$ if there are $n \geq 1$ and $\beta_0, \ldots, \beta_n$ in $(V \cup \Sigma)^*$ s.t.

① $\beta_0 = \alpha$

② $\beta_i \Rightarrow_G \beta_{i+1}$ for each $i = 0, \ldots, n-1$

③ $\beta_n = \beta$.

$$\beta_0 \Rightarrow_G \beta_1 \Rightarrow_G \beta_2 \cdots\cdots \Rightarrow_G \beta_n$$

$$\beta_0 = \alpha, \quad \beta_n = \beta$$

Where, (derivation)

$\alpha \Rightarrow_G \beta$ // α derivates β

if there is a production $A \to \gamma$ in P s.t.

$$\alpha = \alpha_1 A \alpha_2$$

$$\beta = \alpha_1 \gamma \alpha_2 .$$

Example. Consider the following c.f. grammar:

$$S \longrightarrow AB$$

$$A \longrightarrow 0A1 \mid \Lambda$$

$$B \longrightarrow \Lambda$$

$$\alpha = AABA \Rightarrow \beta = AAB0A1$$

$$\Rightarrow AA0A1$$

$$\Rightarrow 0A1A0A1$$

$$\Rightarrow 0A10A1$$

$$\Rightarrow 010A1$$

$$\Rightarrow 0101$$

The language generated by grammar G is defined as $L(G) = \{ w \in \Sigma^* : S \Rightarrow_G^* w \}$.

Let G be: $S \longrightarrow 0S1 \mid \Lambda$

$$L(G) = \{ 0^n 1^n : n \geq 0 \}$$

Let G be: $S \longrightarrow 0S1 \mid 01$

$$L(G) = \{ 0^n 1^n : n \geq 1 \}$$

$\mathcal{L} = \{1^n 0^n 1^m 0^m : n \geq 1, m \geq 0\}$. Give me a gramer to generate the $\mathcal{L}$:

$$S \longrightarrow AB$$

$$A \longrightarrow 1A0 \mid 10$$

$$B \longrightarrow 1B0 \mid \Lambda$$

$\mathcal{L} = \{1^n 0^m 1^m 0^n : n \geq 1, m \geq 0\}$. Give me a gramar to generate the $\mathcal{L}$:

$$S \to 1S0 \mid 1A0$$

$$A \to 0A1 \mid \Lambda$$

$$\mathcal{L} = \{ 1^{2n} 0^{3m} 1^{2m} 0^{3n} : n \geq 1, m \geq 0 \}.$$

Give me a grammar to generate $\mathcal{L}$:

$$S \to 11S000 \mid \cancel{11000}\ 11A000$$

$$A \to 000A11 \mid \Lambda.$$

$$\mathcal{L} = \{ w \# w^r : w \in \{a,b\}^* \}$$

a symbol.

reverse of w.

$(aaba)^r = abaa$

Grammar for $\mathcal{L}$: I look at an example:

aabb # bbaa

$$S \to aSa \mid bSb \mid \#$$

$\mathcal{L} = \{ ww^r : w \in \{a,b\}^* \}$

Grammar? look at example:

aabbbbaa

$S \to aSa \mid bSb \mid \Lambda$

$\mathcal{L} = \{ w \in \{a,b\}^* : w = w^r \}$

Grammar? look at example;

abbba

abba

$S \to aSa \mid bSb \mid a \mid b \mid \Lambda$

Hard:

$$\mathcal{L} = \{ w : w \in \{a,b\}^*, \ \#_a(w) = \#_b(w) \}.$$

number of a's in w

I want a grammar for $\mathcal{L}$:

Look at example.

aababbababba $\in \mathcal{L}$

$\leftarrow$ 6 a's, 6 b's. $\rightarrow$

$\leftarrow$ 5 a's, 6 b's $\rightarrow$

$\leftarrow$ 4 a's, 6 b's $\rightarrow$

Notes:

S: strings with $\#_a = \#_b$,

B: strings with $\#_b = \#_a + 1$;

A: strings with $\#_a = \#_b + 1$;

BB: strings with $\#_b = \#_a + 2$

AA: strings with $\#_a = \#_b + 2$.

$$S \rightarrow aB \mid bA \mid \Lambda$$

$$B \rightarrow aBB \mid bS \mid b$$

$$A \rightarrow bAA \mid aS \mid a$$